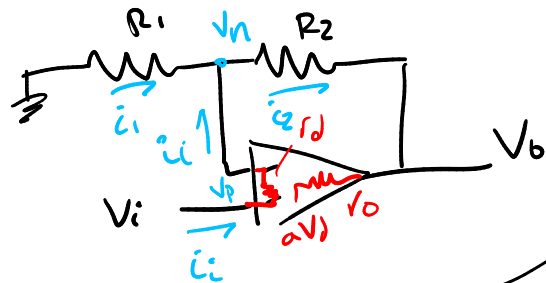


$r_o \neq 0, r_d \neq \infty, a \neq \infty$

find A for non-inv. Amp with using Exact Analysis



$$\begin{cases} V_o = aV_d, & V_d = V_p - V_n = V_i - V_n \\ i_1 + i_i - i_2 = 0 \\ -V_n/R_1 + V_d/r_d - \frac{V_n - V_o}{r_o} = 0 \end{cases}$$

fixed from lecture

$$A = \frac{V_o}{V_i} = \frac{1 + R_2/R_1}{1 + \frac{1}{a} \left[1 + \frac{R_2 + r_o}{R_1} + \frac{R_2 + r_o}{r_d} \right]} + \frac{r_o/r_d}{a + \left[1 + \frac{R_2 + r_o}{R_1} + \frac{R_2 + r_o}{r_d} \right]}$$

$$A = \frac{1 + R_2/R_1}{1 + 1/T} + \dots$$

if $a \rightarrow \infty$

$$A = \frac{(1 + R_2/R_1)}{1 + 1/T} \approx \frac{(1 + R_2/R_1)}{1 + 1/L}$$

Aideal

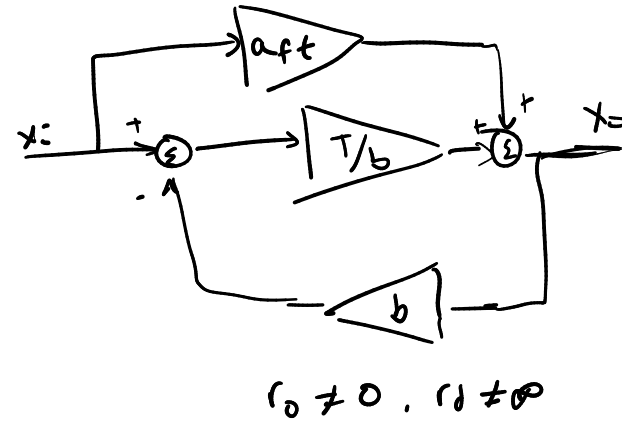
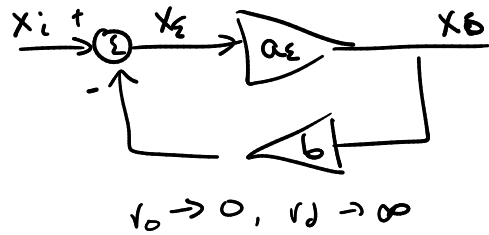
if $a \rightarrow 0$

$A \rightarrow \text{some other value} \rightarrow A_{ft}$

without mess-ups

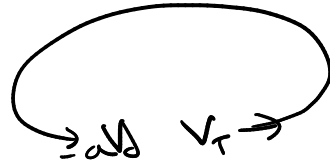
$$A = \frac{A_{ideal}}{1 + 1/T} + \frac{A_{ft}}{1 + T}$$

Return Ratio Analysis $\rightarrow$ Looking for T & a_{ft} $\leftarrow$ feedthrough gain
 $\uparrow$ Return Ratio (very similar to L)



Calculate T (Return Ratio Analysis)

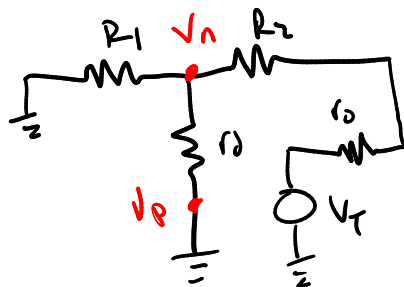
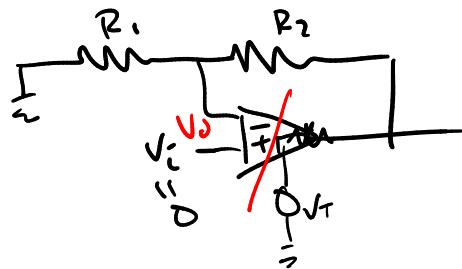
- 1) Set all (independent) sources to 0 ($v_i \rightarrow 0, I_i \rightarrow 0 \dots$)
- 2) Apply a test voltage, V_T , as amp voltage $\rightarrow aV_d \rightarrow V_T$
- 3) $T = \frac{-aV_d}{V_T} \rightarrow$



Aided from $r_d = \infty, r_o = 0, a = \infty$ analysis (use virtual short!)

a_{ft} from $a \rightarrow 0$

RR analysis



$$V_o = V_p - V_n \quad V_p = 0 \rightarrow V_o = -V_n$$

$$T = -\frac{aV_o}{V_T} = +\frac{aV_n}{V_T}$$

$$V_n = R_1 \parallel r_d / (R_1 \parallel r_d + R_2 + r_o) V_T$$

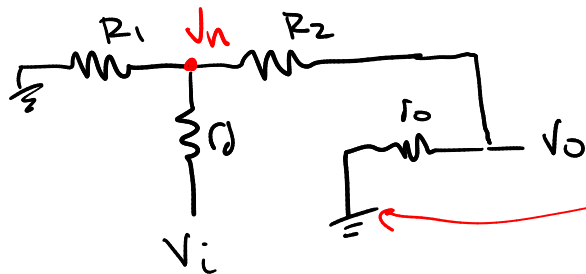
$$V_n = V_T R_1 r_d / [R_1 r_d + R_1 R_2 + R_1 r_o + R_2 r_d + r_o r_d]$$

Clean up $\rightarrow$

$$T = a / [1 + \frac{R_2 + r_o}{R_1} + \frac{R_2 + r_o}{r_d}]$$

$\leftarrow$ still typically large like L , at $f \rightarrow 0$

at $f \rightarrow 0$, $a \rightarrow 0$, $aV_o \rightarrow 0$ Amp output is 0



looking for $a_{ft} = \frac{V_o}{V_i} |_{a \rightarrow 0}$

Assume no load on V_o

$$V_n = V_i \frac{R_1 \parallel (R_2 + r_o)}{R_1 \parallel (R_2 + r_o) + r_d} \quad V_o = V_n \frac{r_o}{R_2 + r_o}$$

$$a_{ft} = \frac{V_o}{V_i} = \frac{r_o / r_d}{1 + \frac{R_2 + r_o}{R_1} + \frac{R_2 + r_o}{r_d}}$$

typically dominates

$$A = \frac{A_{ideal}}{1 + 1/T} + \frac{a_{ft}}{1 + T}$$

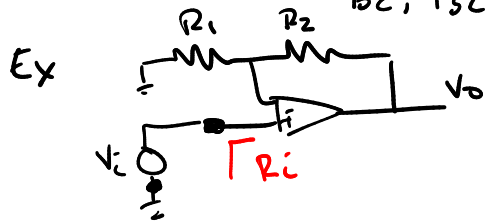
$\checkmark$ likely very small! especially at $f \rightarrow 0$ can probably ignore

Impedances

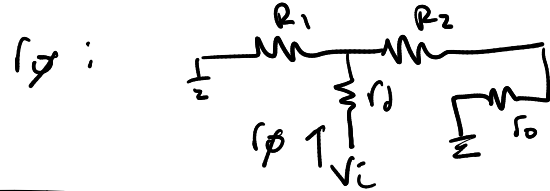
Blackman's (Impedance) formula: $R_x = r_{\phi x} \frac{1 + T_{sc}}{1 + T_{oc}}$

$R_x \rightarrow$ impedance of interest (ex. R_i, R_o)

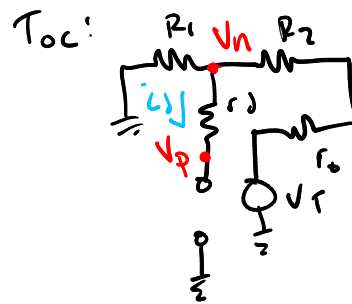
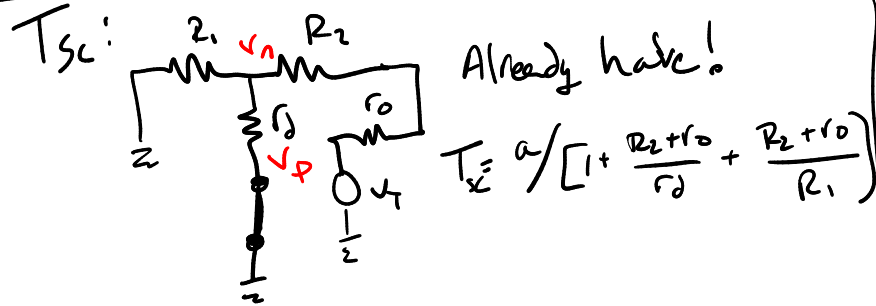
$T_{oc}, T_{sc} \rightarrow$ Return Ratios for if impedance nodes are ^{oc} open or ^{sc} shorted



$r_{\phi} \rightarrow$ impedance at nodes if $a \rightarrow 0$



$$r_{\phi} = r_o + R_1 \parallel (R_2 + r_o) = r_o + \frac{R_1(R_2 + r_o)}{R_1 + R_2 + r_o}$$



$$i_d = 0 \Rightarrow v_n = v_p$$

$$v_d = 0 \quad T = - \frac{a v_d}{v_T} = 0$$

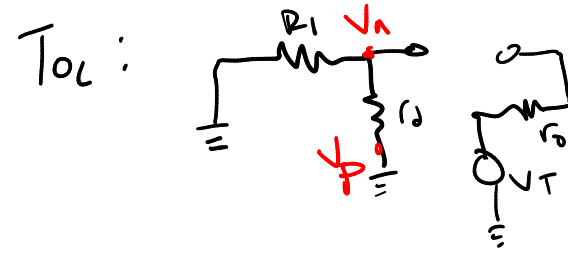
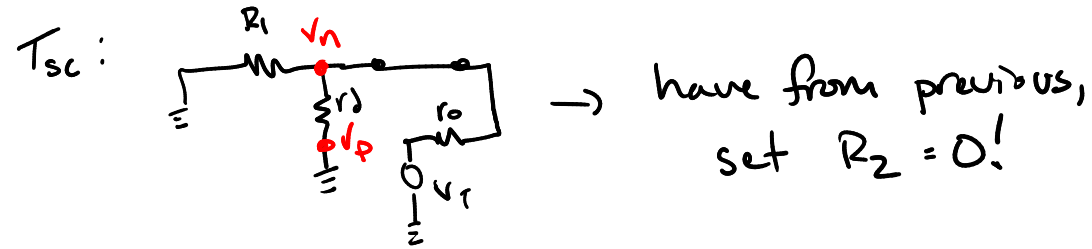
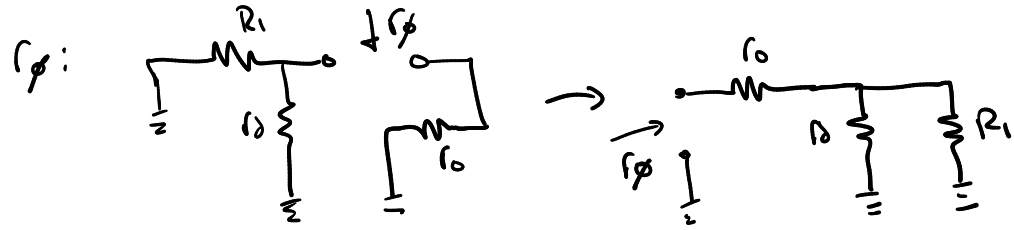
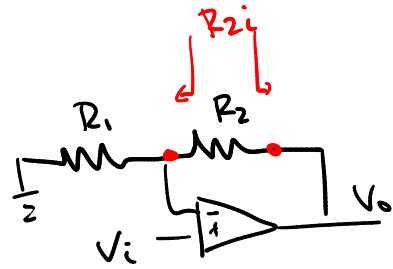
$$T_{oc} = 0$$

Assemble all: $R_i = \left[r_o + \frac{R_1(R_2 + r_o)}{R_1 + R_2 + r_o} \right] \left[1 + a / \left[1 + \frac{R_2 + r_o}{r_o} + \frac{R_2 + r_o}{R_1} \right] \right]$

should be able to verify $R_i = r_o(1 + L)$ if $r_o \rightarrow 0$ ($r_o \rightarrow \infty$)

Blackman's formula

Structural example for another resistance ...

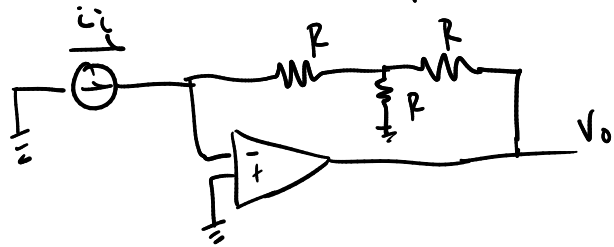


No connection to V_n , $V_p + V_n = V_p = 0$

$$V_d = 0$$

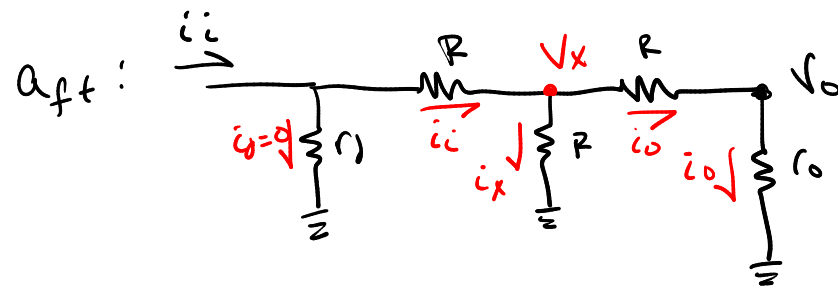
$$\therefore T_{oc} = 0$$

Return Ratio example #2



$$A = \frac{V_o}{i_i} = ? \left[\frac{V}{A} \right]$$

$$a_{ft} = \frac{V_o}{i_i} \bigg|_{a \rightarrow 0} = \frac{1}{\frac{R+r_o}{Rr_o} + \frac{1}{r_o}} = \frac{Rr_o}{2R+r_o}$$



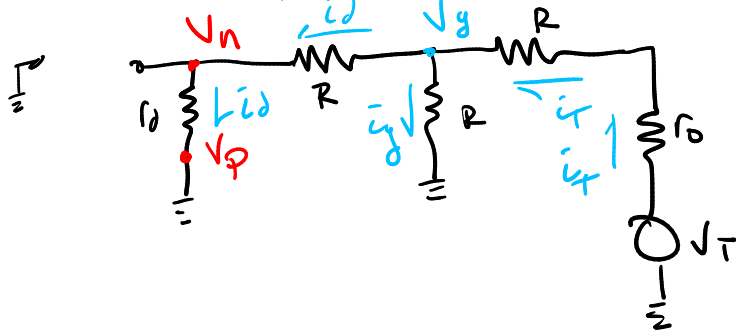
Assumption: $r_o \gg R$, $r_o \rightarrow \infty$, $i_o \rightarrow 0$

$$i_i = i_x + i_o = \frac{V_x}{R} + \frac{V_x}{R+r_o}$$

$$V_o = V_x \frac{r_o}{R+r_o} \rightarrow V_x = V_o \frac{R+r_o}{r_o}$$

$$i_i = \left[\frac{R+r_o}{Rr_o} + \frac{1}{r_o} \right] V_o$$

T: $i_i \rightarrow 0$ (open circuit)



$$V_o = 0 - V_n = -V_n \rightarrow T = -\frac{aV_d}{V_T} = \frac{aV_n}{V_T}$$

$$i_T = i_d + i_g = \frac{V_g}{R+r_d} + \frac{V_g}{R} = \frac{(V_T - V_g)}{R-r_o}$$

$$V_n = V_g \frac{r_d}{R+r_d}$$

$$V_T = \left[\frac{(R+r_o)(R+r_d)}{r_d} \right] \left[\frac{1}{R+r_o} + \frac{1}{R} + \frac{1}{R+r_d} \right] V_n$$

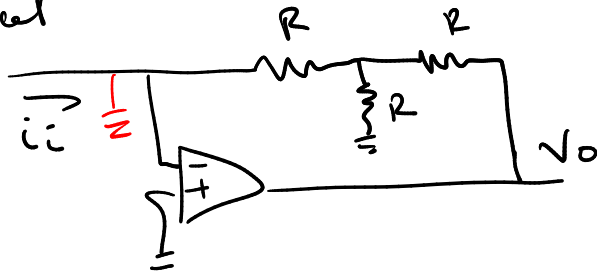
$$T = a / \left[\right]$$

L: $r_d \rightarrow \infty$, $r_o \rightarrow 0$

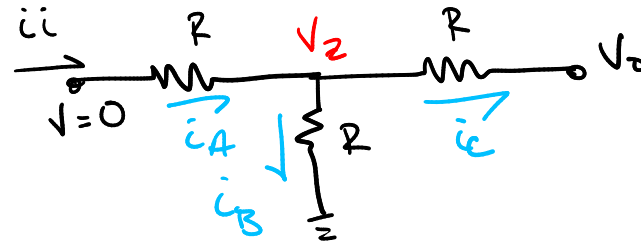
$$L = T \bigg|_{r_d \rightarrow \infty, r_o \rightarrow 0} = a / \left[\frac{R+r_d}{r_d} \left(\frac{1}{R} + \frac{1}{R} + \frac{1}{\infty} \right) \right]$$

$$L = \frac{a}{2} \text{ if } a_2 = a, \text{ then } b = \frac{1}{2}$$

Aideal



$$r_o \rightarrow \infty, r_o \rightarrow 0, a \rightarrow \infty$$



No load on V_o

$$\therefore i_c = 0$$

$$i_A = i_B$$

$$V_2 = V_o$$

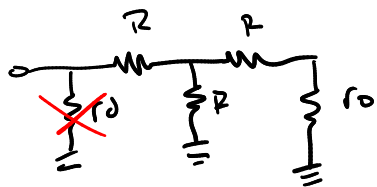
$$\frac{0 - V_2}{R} = i_i$$

$$V_o = V_2 = -R i_i$$

$$\frac{V_o}{i_i} = -R$$

Calculate R_i :

r_o :

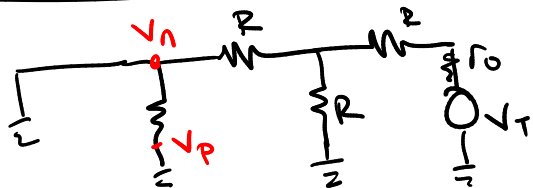


$$r_o \gg R \quad r_o \rightarrow \infty$$

$$r_o = R + R \parallel (R + r_o) = \frac{R(R + r_o)}{2R + r_o} + R$$

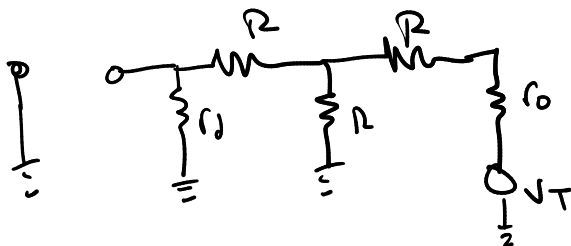
$$\text{if } r_o = 0 \quad r_o = \frac{3R}{2}$$

T_{sc} :



$$V_n = V_p = 0 \quad T_{sc} = 0 \quad (V_d = 0)$$

T_{oc} :

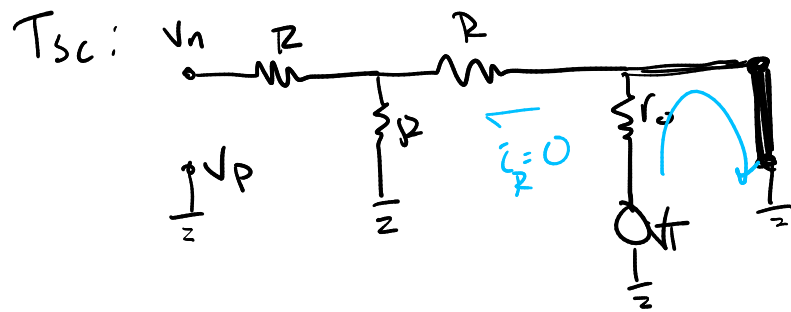
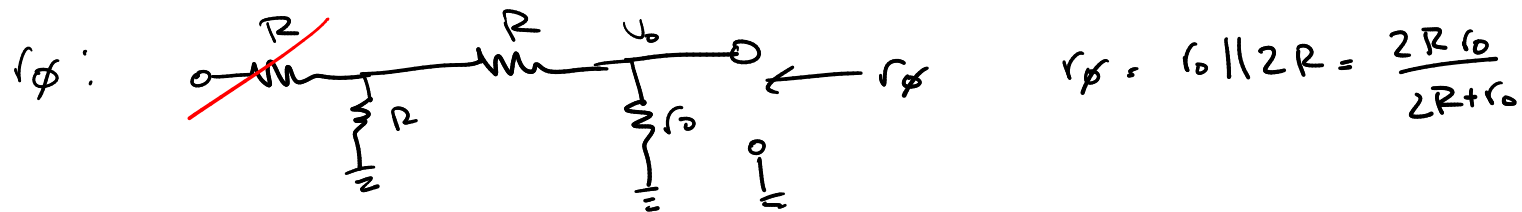


Same as before. $T_{oc} \approx \frac{\alpha}{2}$
(full expression here....)

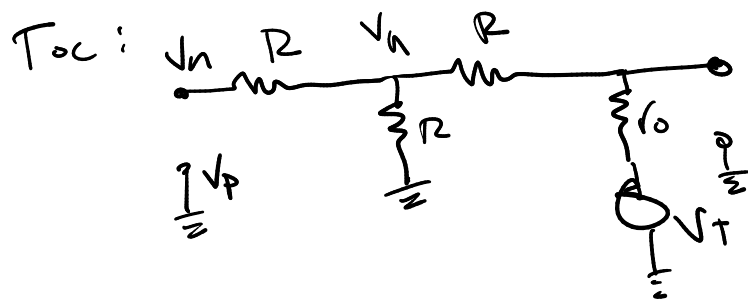
$$R_i = r_o \frac{1 + T_{sc}}{1 + T_{oc}} = \left[\frac{R(R + r_o)}{2R + r_o} + R \right] \left[1 + \frac{R r_o \alpha}{(3R^2 + 2R r_o + 2R r_o + r_o r_o)} \right]$$

$$r_o \rightarrow \infty, r_o \rightarrow 0 \quad R_i = \frac{3R}{2} \left[1 + \frac{\alpha}{2} \right]$$

Output Impedance: R_o (assume $r_d \rightarrow \infty$)



$V_n = 0 \quad V_d = 0 \quad T_{sc} = 0$



$V_n = V_T \frac{R}{2R+r_o} \quad V_d = -V_n$

$T_{oc} = -a \frac{V_d}{K} = \frac{aR}{2R+r_o}$

$R_o = r_o \frac{1+T_{sc}}{1+T_{oc}} = \frac{2Rr_o}{2R+r_o} / 1 + \frac{aR}{2R+r_o}$

if $r_o \rightarrow \infty$ $R_o = \frac{2Rr_o}{2R} / 1 + a/2$

$R_o = r_o / 1 + a/2 \approx r_o / 1 + T$